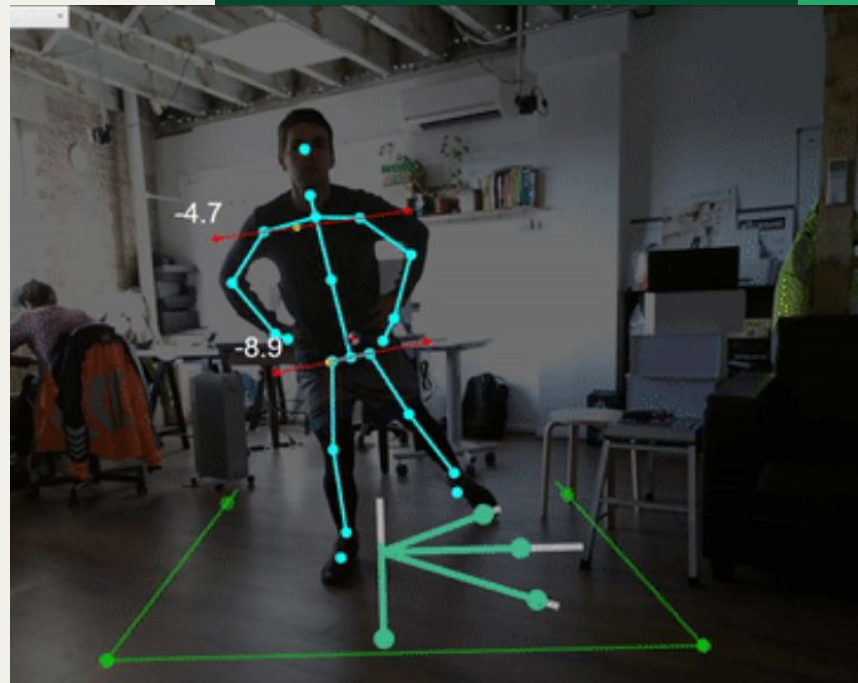


Predicting Activity Levels with LSTM for mHealth Applications: A ML approach for Sensor Based Activity Classification

Datathon 4 By Team 6: Yu Qing (Jenny)
Huang, Marya Zaidi,
Batool AlMousawi



Predicting Activity Levels with LSTM for mHealth Applications

Problem: 1.8 billion adults worldwide did not meet recommended physical activity levels in 2022 (WHO)

Objective: Develop a machine learning-based solution using wearable sensor data to classify physical activity levels, aiming to identify individuals who require targeted health interventions.

Dataset: Sunnybrook Health Sciences Center (13 activity classes from sedentary to vigorous).

Clinical Impact: Potential use in real-time activity tracking & intervention targeting, but needs better explainability & diverse datasets.

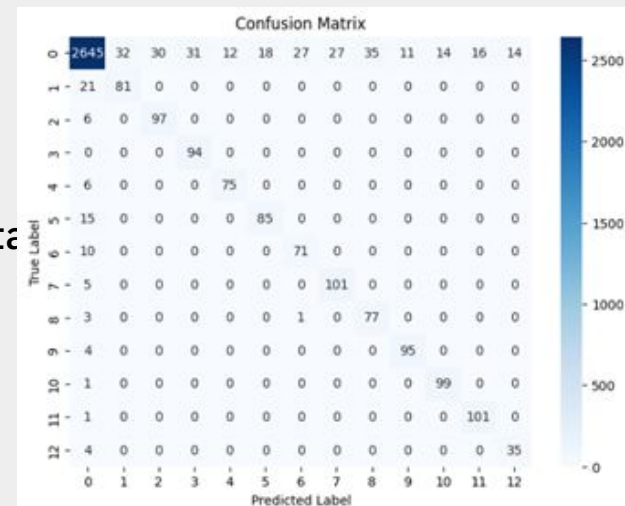
Data Preprocessing & Feature Engineering

- Data normalization for consistent scaling.
- Sliding window segmentation: 2-second windows, 50 samples/sec, stride of 2 seconds.
- Extracted features: Root Mean Square (RMS), Signal Magnitude Area (SMA), Jerk, and Variance.

Methodology & Key Findings

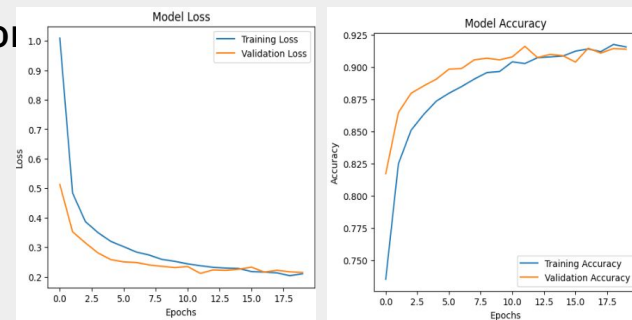
Modeling Approach

- **LSTM Neural Network:**
 - Chosen for sequential pattern recognition in time-series data.
 - Softmax activation for multi-class classification.
- **Training & Validation:**
 - 80:20 train-test split.
 - 20 epochs, batch size = 32.
 - ADAM optimizer, categorical cross-entropy loss.



Key Findings

- Overall accuracy: 91%.
- Performance variability across classes due to class imbalance:
 - Best performance in sedentary (class 0, F1=0.94) and vigorous activities.
 - Moderate performance in light activities (due to class imbalance)



From Model to Real-World Application: What's Next?

Challenges:

Explainability: Deep learning “black box” hinders clinical insights.

Class Imbalance: Majority class skewed results → SMOTE or weighted loss functions can help.

Generalizability: Data from 9 young, healthy participants → Need diverse datasets.

Next Steps:

- Feature importance analysis (e.g., SHAP) for better interpretability.
- Fine-tune architecture (e.g., bidirectional LSTM, attention mechanisms).
- Test on broader populations for real-world robustness.

	precision	recall	f1-score	support
0	0.97	0.91	0.94	2912
1	0.72	0.79	0.75	102
2	0.76	0.94	0.84	103
3	0.75	1.00	0.86	94
4	0.86	0.93	0.89	81
5	0.83	0.85	0.84	100
6	0.72	0.88	0.79	81
7	0.79	0.95	0.86	106
8	0.69	0.95	0.80	81
9	0.90	0.96	0.93	99
10	0.88	0.99	0.93	100
11	0.86	0.99	0.92	102
12	0.71	0.90	0.80	39
accuracy			0.91	4000
macro avg	0.80	0.93	0.86	4000
weighted avg	0.92	0.91	0.92	4000